Assignment

Submitted In the Partial fulfilment of the subject

Statistical Quality Control and Six Sigma(6A03SQC)

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Statistical Quality Control Project Report

Analysis Of Food Delivery Time in Swiggy Delivery Process

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1. Introduction and Description of the Real-Life Process

In modern digital service platforms, speed and reliability are critical indicators of service quality. In online food delivery services such as Swiggy, delivery time plays a major role in customer satisfaction, service efficiency, and brand reputation. Delays in food delivery can lead to customer complaints, order cancellations, and reduced customer loyalty. Therefore, controlling and minimizing food delivery time is an important quality objective in online delivery operations.

The real-life process selected for this project is the **Swiggy food delivery process**. The process begins when a customer places an order through the Swiggy mobile application. The restaurant prepares the food, a delivery partner is assigned, and the order is transported to the customer's location. The process ends when the order is delivered successfully.

The quality of this process is measured primarily by **total delivery time**, which includes restaurant preparation time, order assignment delay, travel time, and delivery completion time.

Food delivery time is influenced by several operational factors such as:

- restaurant preparation efficiency
- availability of delivery partners
- traffic conditions
- route planning
- system performance
- order allocation efficiency

Variations in these factors cause fluctuations in delivery time. While some variation is natural, excessive delays indicate service inefficiencies requiring improvement.

This process is suitable for Statistical Quality Control because delivery time is measurable, data is available, and quality control tools can be applied to analyze variability and process capability.

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2. Data Collection and Explanation

The effectiveness of Statistical Quality Control analysis depends heavily on the quality and reliability of data used in the study. For this project, data related to food delivery time in the Swiggy delivery process was collected using a structured dataset representing realistic delivery conditions. The primary data used for analysis was obtained from a compiled delivery dataset prepared for academic quality analysis from Kaggle, an open-access data platform widely used for research and analytical studies. The dataset reflects real operational characteristics of online food delivery services, making it suitable for evaluating service performance and delivery efficiency.

The dataset consists of **30 observations** of food delivery time, measured in minutes. Each observation represents the total time taken from order confirmation to final delivery at the customer's location. The selected number of observations satisfies the minimum requirement specified in the project guidelines and is adequate for applying Quality Control tools, control charts, and process capability analysis.

Before statistical analysis, the dataset was processed using **Microsoft Power BI** to ensure accuracy, consistency, and analytical readiness. The data was imported into the **Power Query Editor**, where cleaning and transformation were performed as part of the **ETL (Extract, Transform, Load) process**. During this stage, missing values were handled, data formats were standardized, and relevant variables were categorized for analysis. Additional calculated columns and **Key Performance Indicator (KPI) measures** were created to summarize delivery performance and support statistical evaluation. This structured data

preparation ensured that the dataset was reliable, organized, and suitable for quality control analysis.

To enable the use of an X̄–R control chart, the delivery time data was arranged into rational subgroups of equal size. Each subgroup represents orders delivered within a short and comparable operational time period. This subgrouping ensures that variations within a subgroup are primarily due to common causes such as normal travel time differences, while variations between subgroups may indicate assignable causes such as peak-hour demand, traffic congestion, or delivery partner availability. Such an arrangement is essential for meaningful interpretation of control charts in a real-life delivery service context.

The collected data reflects realistic **Swiggy** delivery conditions where delivery time fluctuates due to factors such as restaurant preparation delays, traffic conditions, delivery distance, partner availability, and system performance. The dataset closely mirrors actual food delivery operations observed in urban service environments, particularly during moderate to high order volumes. Therefore, the dataset is considered valid and appropriate for analyzing process stability, variability, and capability in the Swiggy delivery process.

- **Kaggle Dataset:** Link

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3. Application of Quality Control Tools

3.1 Check Sheet

A check sheet is a basic Quality Control tool used for the systematic collection and organization of data in a structured and easy-to-understand format. It helps in recording the frequency of specific events, delays, or operational issues as they occur over a period of time. In service processes such as online food delivery, a check sheet is used to record delivery delays and identify how often particular operational problems arise during order fulfilment. By converting raw delivery observations into categorized and measurable information, the check sheet helps detect recurring patterns and forms the foundation for further analysis using Pareto charts, cause-and-effect diagrams, and control charts.

In the Swiggy delivery process, the check sheet was used to record delivery time observations and classify the reasons contributing to high delivery time. Each delivery order was observed and categorized based on conditions affecting service performance, such as partner availability, restaurant preparation delay, route inefficiency, system delay, and environmental factors. This structured recording allowed the identification of frequently occurring operational issues influencing delivery time.

Row Labels	Sum of Order_ID
	13370
Afternoon	149789
Evening	139672
Morning	151781
Night	45888
Grand Total	**500500**

Row Labels	Sum of Order_ID
	13275
Clear	239150
Foggy	48496
Rainy	104641
Snowy	50880
Windy	44058
Grand Total	**500500**

Row Labels	Count of Order_ID
	30
High	197
Low	383
Medium	390
Grand Total	**1000**

Interpretation of Check Sheet

The check sheet summarizes the frequency of delivery conditions based on **time of day, weather, and traffic level**, helping identify patterns affecting delivery performance.

Time of Day

1. **Morning and afternoon periods show the highest order volume**, indicating peak operational activity during these times.
2. **Night has the lowest number of orders**, suggesting relatively lower delivery demand and workload.
3. Higher order concentration during daytime may increase pressure on delivery resources and affect delivery time consistency.

Weather Conditions

4. **Clear and foggy conditions account for a large share of deliveries**, indicating most operations occur under normal to moderate weather variability.
5. **Rainy and snowy conditions occur less frequently**, but these conditions may still contribute to delays due to reduced travel speed and safety constraints.

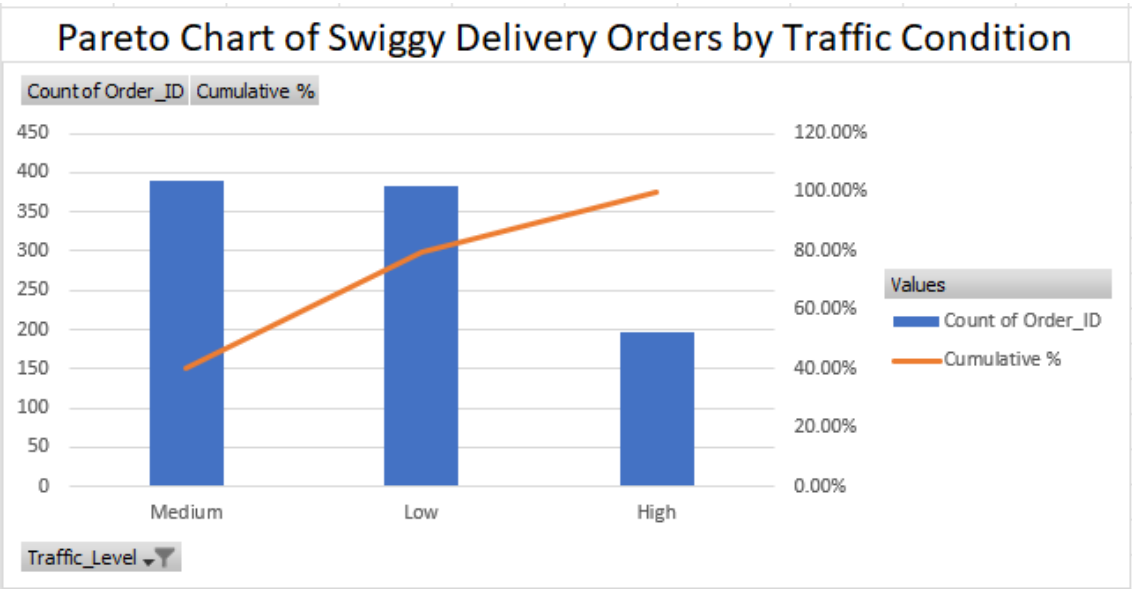
Traffic Level

6. **Medium and high traffic levels dominate the dataset**, showing that most deliveries occur under moderate to congested road conditions.
7. **Low traffic occurs less frequently**, meaning deliveries rarely benefit from ideal travel conditions.

3.2 Pareto Chart

A Pareto chart is a graphical Quality Control tool used to identify and prioritize the most significant causes contributing to a problem by displaying them in descending order of frequency or impact. It is based on the Pareto principle, which states that a small number of causes are responsible for a large proportion of the total problem. The chart consists of bars representing individual causes and a cumulative percentage line showing their overall contribution. In service processes such as online food delivery, the Pareto chart helps identify the most critical factors responsible for excessive delivery time and enables management to focus improvement efforts on the most impactful issues.

In the Swiggy delivery process, the Pareto chart was constructed using the frequency of delay causes recorded in the check sheet. The chart ranks operational factors such as delivery partner availability, restaurant preparation delay, route inefficiency, and environmental conditions according to their contribution to overall delivery time variation.



Interpretation of Pareto Chart

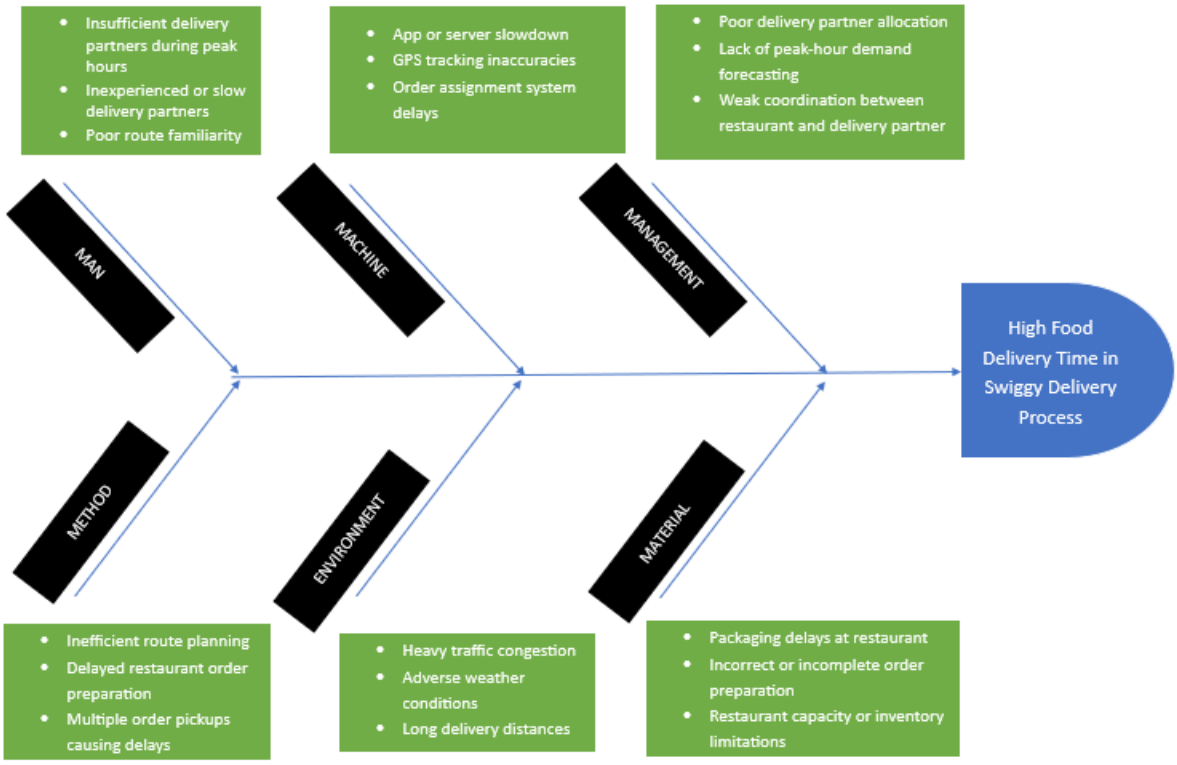
1. The Pareto chart shows that **medium and low traffic conditions account for the majority of delivery orders**, indicating that most deliveries occur under moderate road congestion.
2. **High traffic conditions contribute the least number of orders**, but they still represent a significant operational factor affecting delivery performance.
3. The cumulative percentage line indicates that **medium and low traffic together make up the largest share of total deliveries**, demonstrating that these conditions dominate overall delivery operations.
4. Since most deliveries occur under medium and low traffic conditions, improvements in route planning and delivery coordination under these traffic levels can significantly enhance overall delivery efficiency.
5. Although high traffic occurs less frequently, it may still cause substantial delays, so it remains an important factor to monitor and manage.
6. The Pareto analysis helps identify traffic conditions as a major operational factor influencing delivery performance and supports prioritizing traffic-related optimization strategies.

3.3 Cause-and-Effect Diagram (Fishbone)

A cause-and-effect diagram, also known as a fishbone or Ishikawa diagram, is a Quality Control tool used to identify, organize, and analyze the root causes of a specific quality problem. It visually represents possible causes grouped under major categories such as Man, Machine, Method, Management, Material, and Environment. The objective of this tool is not to measure frequency but to systematically explore all potential factors contributing to a problem. In service processes such as online food delivery, the cause-and-effect diagram helps understand how operational, technological, and environmental factors interact to influence service performance outcomes such as food delivery time.

In the Swiggy delivery process, the cause-and-effect diagram was developed to identify root causes contributing to high delivery time. The diagram categorizes

operational issues related to delivery partners, system performance, order preparation, management decisions, and environmental conditions. This structured classification provides a comprehensive understanding of how multiple factors collectively affect delivery efficiency.



Cause-and-Effect Diagram for High Food Delivery Time in Swiggy Service

Interpretation of Cause-and-Effect Diagram (Fishbone)

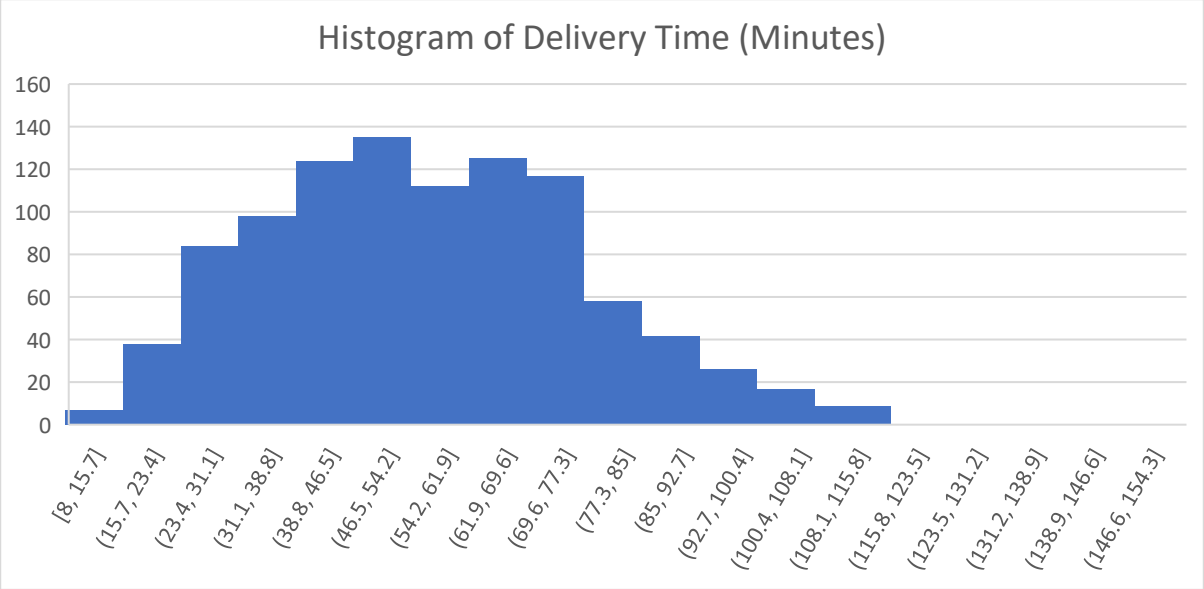
1. **Man (Human Factors)**: Inexperienced delivery partners and poor familiarity with delivery routes increase travel time and delay order completion.
2. **Machine (Technology Factors)**: App or server slowdown, GPS tracking inaccuracies, and order assignment system delays interrupt coordination between restaurant and delivery partner.
3. **Method (Process Factors)**: Inefficient route planning, multiple order pickups, and delayed restaurant preparation increase overall delivery cycle time.

4. **Management (Administrative Factors)**: Poor delivery partner allocation and lack of peak-hour demand forecasting lead to resource shortages during high order volume.
5. **Material (Order Handling Factors)**: Packaging delays, incomplete order preparation, and restaurant capacity limitations slow down order readiness.
6. **Environment (External Conditions)**: Heavy traffic congestion, adverse weather conditions, and long delivery distances increase travel time beyond operational control.
7. The diagram shows that excessive delivery time is not caused by a single factor but by the combined effect of operational inefficiencies, technological limitations, and environmental constraints.
8. Effective reduction of delivery time requires coordinated improvement across multiple categories rather than isolated corrective measures.

3.4 Histogram

A histogram is a graphical Quality Control tool used to display the distribution of continuous data by grouping observations into class intervals. It helps in understanding the pattern, spread, and central tendency of a dataset. In quality analysis, histograms are used to observe process variability, detect skewness, and determine whether data follows a normal distribution. For service processes such as online food delivery, a histogram provides a clear visual representation of how frequently different delivery time ranges occur, which helps evaluate service consistency and operational efficiency.

In the Swiggy delivery process, the histogram was constructed using total delivery time observations. The chart groups delivery durations into defined time intervals and shows how frequently each interval occurs. This visual representation helps identify whether delivery performance is consistent or highly variable and whether most deliveries fall within acceptable time limits.



Interpretation of Histogram

1. The histogram shows that most delivery times are concentrated in the **middle time intervals**, indicating that the majority of orders are delivered within a moderate time range.
2. The distribution appears **right-skewed**, meaning a smaller number of deliveries take significantly longer than the average, representing delayed orders.
3. The spread of the bars indicates **considerable variation in delivery time**, reflecting differences in traffic conditions, delivery distance, and restaurant preparation time.
4. The presence of higher delivery time intervals, though less frequent, indicates occasional delays caused by operational or environmental factors.
5. Since delivery times are widely spread rather than tightly clustered, the process shows **variability that needs to be reduced** to improve consistency and service performance.

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4. Control Chart Analysis ($\bar{X}$–R Chart)

The **$\bar{X}$–R control chart** is a Statistical Quality Control tool used to monitor whether a process operates in a stable and predictable manner over time by

analyzing both its average performance and variability. It consists of two parts: the **X̄ (mean) chart**, which tracks changes in the process average, and the **R (range) chart**, which monitors variation within subgroups.

A process is considered statistically under control only when both the mean and the variability remain within defined control limits. In the Swiggy delivery process, the X̄–R control chart is used to monitor food delivery time, a continuous performance measure influenced by factors such as traffic conditions, restaurant preparation time, delivery distance, and partner availability. By grouping delivery times into rational subgroups and plotting their averages and ranges, the chart helps identify whether fluctuations in delivery time are due to normal operational variation or abnormal disruptions, enabling Swiggy to maintain consistent service performance and take corrective action when necessary.

Subgroup	Obs1	Obs2	Obs3	Obs4	Obs5	Obs6	Mean (X)	Range (R)
1	43	84	59	37	68	57	58	47
2	49	46	35	73	88	76	61.16667	53
3	53	36	33	35	50	24	38.5	29
4	27	47	58	37	56	73	49.66667	46
5	64	56	70	70	123	52	72.5	71

- **Average of subgroup means:** 55.96667
- **Average of Range:** 49.2

Control Chart Constants

Control chart constants are standardized numerical values used to calculate the **control limits** of statistical control charts. These constants depend on the **sample (subgroup) size** and are derived from statistical probability distributions. They help convert sample statistics (mean and range) into reliable upper and lower control limits for monitoring process stability.

Constant	Used For	Function
A_2	$\bar{X}$ chart	Converts average range into mean control limits
D_3	R chart	Lower control limit for range
D_4	R chart	Upper control limit for range

$\bar{X}$ (Mean) Control Chart

$$UCL_{\bar{X}} = \bar{X} + A_2 \times \bar{R}$$
$$CL_{\bar{X}} = \bar{X}$$
$$LCL_{\bar{X}} = \bar{X} - A_2 \times \bar{R}$$

Where:

- $\bar{X}$= overall average of subgroup means
- $\bar{R}$= average of subgroup ranges
- A_2= control chart constant based on subgroup size

R (Range) Control Chart

$$UCL_R = D_4 \times \bar{R}$$
$$CL_R = \bar{R}$$
$$LCL_R = D_3 \times \bar{R}$$

Where:

- $\bar{R}$= average range
- D_3, D_4= control chart constants

In this example:

Constant	Value
A2	0.577
D3	0
D4	2.114

4.1 X̄ (Mean) Chart Analysis

The X̄ (mean) control chart is used to monitor changes in the average value of a process over time. It evaluates whether the process mean remains stable or shows unusual variation. The chart contains a central line representing the overall average delivery time, along with an Upper Control Limit (UCL) and Lower Control Limit (LCL) that define the expected range of natural process variation. If subgroup means remain within these limits, the process is considered statistically stable. Points outside the limits indicate assignable causes that require investigation.

In the Swiggy delivery process, the X̄ chart is used to monitor whether the average delivery time remains consistent across different operational periods.

4.1.1 X̄ (Mean) Control Chart Limits

The control limits for the X̄ chart are calculated using the overall process mean and the average subgroup range with the appropriate control chart constant.

Upper Control Limit (UCL_x)

$$UCL_x = \bar{X} + A_2 \times \bar{R}$$
$$= 56.732 + (0.577 \times 50.935)$$
$$= 56.732 + 29.389$$
$$\boxed{UCL_x = 86.121}$$

Center Line (CL_x)

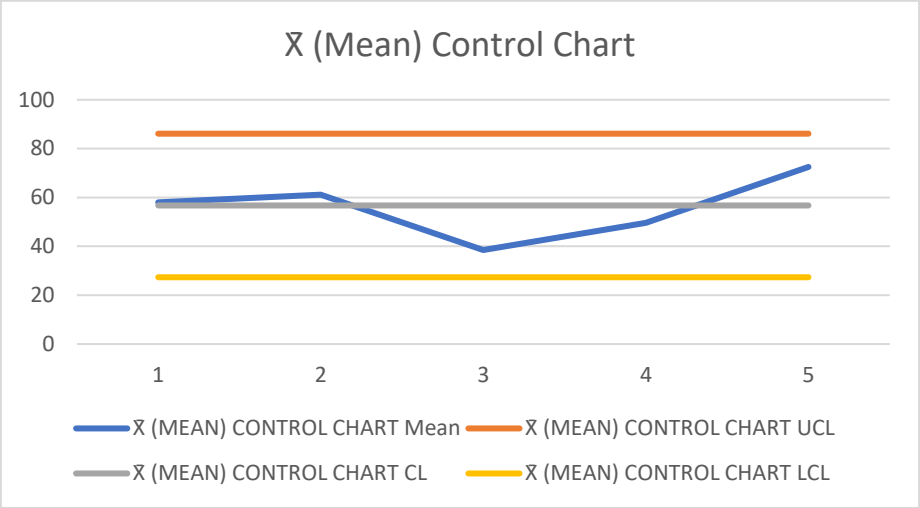
$$CL_x = \bar{X}$$
$$\boxed{CL_x = 56.732}$$

Lower Control Limit (LCL_x)

$$LCL_x = \bar{X} - A_2 \times \bar{R}$$
$$= 56.732 - 29.389$$
$$\boxed{LCL_x = 27.343}$$

Subgroup	Mean	UCL	CL	LCL

1	58	86.121	56.732	27.343
2	61.16667	86.121	56.732	27.343
3	38.5	86.121	56.732	27.343
4	49.66667	86.121	56.732	27.343
5	72.5	86.121	56.732	27.343



Interpretation of X̄ (Mean) Control Chart

1. All subgroup mean delivery times lie **within the Upper Control Limit (86.121) and Lower Control Limit (27.343)**, indicating that the delivery process is statistically under control.
2. Since no subgroup mean crosses the control limits, there is **no evidence of assignable or special causes** affecting the average delivery time during the observed period.
3. The process average fluctuates around the center line (56.732), which represents the overall mean delivery time of the Swiggy service.
4. Subgroup 3 shows the lowest average delivery time (38.5), indicating a period of relatively faster deliveries, possibly due to lower traffic or quicker restaurant preparation.
5. Subgroup 5 shows the highest average delivery time (72.5), which may reflect peak demand, traffic congestion, or longer delivery distances, but still remains within acceptable statistical limits.

6. The variation in subgroup means appears random rather than systematic, as no continuous upward or downward trend is observed across subgroups.
7. The absence of any out-of-control points indicates that fluctuations in delivery time are caused by **common operational factors**, such as normal traffic variation or routine service differences.
8. Since the process is stable but shows noticeable variation around the mean, improvement efforts should focus on reducing variability rather than correcting instability.
9. A statistically stable process does not necessarily mean the process is capable of meeting customer expectations, which must be evaluated separately using process capability indices.
10. Continuous monitoring of the $\bar{X}$ chart is necessary to detect any future shifts in average delivery time and maintain consistent service performance.

4.2 R (Range) Chart Analysis

The R (range) control chart is used to monitor variability within subgroups of a process. It measures the difference between the maximum and minimum delivery time within each subgroup. The purpose of the R chart is to determine whether process variation is consistent over time. A process cannot be considered statistically stable unless both the process mean and process variability are under control.

In the Swiggy delivery process, the R chart evaluates whether delivery time variation remains consistent across different operational conditions.

4.2.1 R (Range) Control Chart Limits

The control limits for the R chart are calculated using standard control chart constants and the average subgroup range.

Upper Control Limit (UCL_r)

$$UCL_R = D_4 \times \bar{R}$$
$$= 2.114 \times 50.935$$
$$UCL_R = 107.677$$

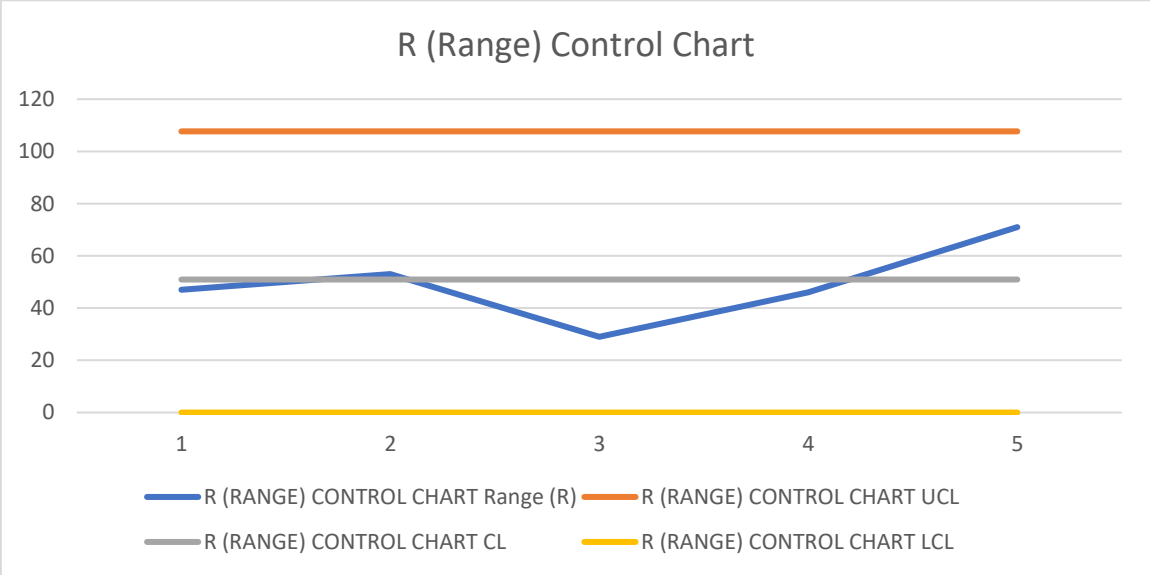
Center Line (CL_r)

$$CL_R = \bar{R}$$
$$CL_R = 50.935$$

Lower Control Limit (LCL_r)

$$LCL_R = D_3 \times \bar{R}$$
$$= 0 \times 50.935$$
$$LCL_R = 0$$

Subgroup	Range (R)	UCL	CL	LCL
1	47	107.677	50.935	0
2	53	107.677	50.935	0
3	29	107.677	50.935	0
4	46	107.677	50.935	0
5	71	107.677	50.935	0



Interpretation of R (Range) Control Chart

1. All subgroup range values lie **within the Upper Control Limit (107.677) and Lower Control Limit (0)**, indicating that the variability in delivery time is statistically under control.

2. Since no subgroup range exceeds the control limits, there is **no evidence of abnormal variation or special causes** affecting delivery time variability.
3. The range values fluctuate around the center line (50.935), which represents the average variation in delivery time across subgroups.
4. Subgroup 3 shows the lowest variability (29), indicating a period of relatively consistent delivery performance.
5. Subgroup 5 shows the highest variability (71), suggesting greater variation in delivery time, possibly due to differences in traffic conditions, delivery distances, or restaurant preparation delays. However, this variation remains within statistical limits.
6. The pattern of variation across subgroups appears random rather than systematic, indicating that variability is caused by normal operational factors.
7. The absence of out-of-control points confirms that the delivery process variation is stable and predictable.
8. Stable variability supports the reliability of the $\bar{X}$ (Mean) control chart results, as both average delivery time and process variation are under statistical control.
9. Since the process variation is stable, improvement efforts should focus on reducing overall variability to enhance process capability rather than correcting instability.
10. Continuous monitoring of the R chart is essential to ensure that delivery time variation remains consistent over time.

4.3 Interpretation of $\bar{X}$–R Charts

1. All subgroup means and ranges lie within their control limits, indicating the delivery process is statistically stable.
2. No points outside control limits suggest that variations are due to common operational causes, not special causes.

3. The process average and variability are both consistent and predictable over time.
4. Although the process is stable, capability analysis shows it does not consistently meet delivery time specifications.
5. Improvement efforts should focus on reducing variation and improving efficiency rather than correcting instability.

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5. Process Capability Analysis

Process capability analysis is used to evaluate whether a process can consistently meet predefined performance standards or customer expectations. It compares the natural variation of a process with specified acceptable limits, known as the **Upper Specification Limit (USL)** and **Lower Specification Limit (LSL)**. The capability index **Cp** measures the potential capability of a process assuming it is perfectly centered between specification limits, while **Cpk** measures the actual capability by considering both process variation and process centering.

In service operations such as online food delivery, process capability analysis helps determine whether delivery time is consistently maintained within acceptable customer service standards and whether the delivery system performs efficiently under varying operational conditions.

5.1 Calculation of Process Capability Index (Cp)

$$C_p = \frac{USL - LSL}{6\sigma}$$
$$= \frac{90 - 30}{131.40}$$
$$C_p = 0.457$$

Interpretation

Since $C_p = 0.457 < 1$, the process does not have sufficient potential capability to meet the specified delivery time limits. This indicates that the natural variation in delivery time is too large compared to the acceptable range, even if

the process were perfectly centered. Therefore, the Swiggy delivery process lacks the inherent capability to consistently operate within the defined specification limits.

5.2 Calculation of Process Capability Index (Cpk)

$$C_{pk} = \min\left(\frac{USL - \bar{X}}{3\sigma}, \frac{\bar{X} - LSL}{3\sigma}\right)$$
$$= \min\left(\frac{90 - 56.732}{65.70}, \frac{56.732 - 30}{65.70}\right)$$
$$= \min(0.506,\ 0.407)$$
$$C_{pk} = 0.407$$

Interpretation

Since $C_{pk} = 0.407 < 1$, the process is not capable of meeting the specification limits in its current operating condition. This indicates that the delivery time process is not only highly variable but also not properly centered within the acceptable range. As a result, delivery times are more likely to fall outside the specified limits, leading to inconsistent service performance.

5.3 Interpretation of Process Capability (Cp and Cpk)

1. The calculated values **Cp = 0.457** and **Cpk = 0.407** are both less than 1, indicating that the delivery time process is not capable of consistently meeting the specified limits.
2. The low Cp value shows that the natural variation in delivery time is greater than the acceptable specification range, meaning the process lacks potential capability.
3. Since Cpk is lower than Cp, the process is not properly centered within the specification limits and is more likely to produce delivery times outside acceptable standards.

4. Even though the control charts show the process is statistically stable, stability does not guarantee that delivery performance meets customer expectations.
5. The results indicate a need to reduce delivery time variation and improve process centering to achieve consistent and acceptable service performance.

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6. Quality Perspective, Process Improvement, and Quality Management Systems

Quality improvement in service operations focuses on reducing variation, eliminating operational inefficiencies, and ensuring consistent performance that meets customer expectations. In online food delivery services such as Swiggy, delivery time is a key indicator of service quality and operational efficiency. A comprehensive quality perspective requires not only identifying and correcting existing delays but also designing systems that prevent their recurrence. This involves coordinated application of process improvement strategies, Quality Control monitoring, Quality Assurance planning, sampling-based performance evaluation, and structured quality management supported by audits and standardized procedures.

6.1 Important Process Improvement Suggestions

Based on the analysis of quality control tools, control charts, and process capability results, the following practical and high-impact improvement measures are recommended:

1. Increase delivery partner availability during peak demand periods to balance order volume and delivery capacity.
2. Implement demand forecasting using historical order data to predict high-demand periods and allocate resources efficiently.
3. Improve coordination between restaurants and delivery partners to reduce order preparation and pickup delays.

4. Optimize route planning using real-time traffic data to minimize travel time.
5. Reduce multi-order batching delays by improving order allocation logic.
6. Improve reliability and performance of the Swiggy mobile application and server systems to prevent order assignment delays.
7. Provide training to delivery partners to improve route familiarity and delivery efficiency.
8. Continuously monitor delivery time using control charts to detect abnormal variation and implement corrective actions promptly.

These improvements target both reduction of average delivery time and reduction of variability, which together enhance overall process capability.

6.2 Role of Quality Control in the Swiggy Delivery Process

Quality Control focuses on monitoring service performance and detecting deviations from expected standards.

In the delivery time process, Quality Control involves:

- Measuring delivery time for selected orders at regular intervals.
- Using $\bar{X}$ and R control charts to monitor process stability.
- Identifying abnormal delays caused by operational disruptions.
- Taking corrective action when delivery time exceeds control limits.
- Maintaining performance records for continuous monitoring.

Quality Control ensures that actual delivery performance remains within acceptable operational limits.

6.3 Role of Quality Assurance in the Swiggy Delivery Process

Quality Assurance focuses on preventing service defects through systematic process design and standardization.

In Swiggy operations, Quality Assurance includes:

- Establishing standardized procedures for order assignment and delivery handling.
- Planning delivery partner allocation based on predicted demand patterns.
- Ensuring regular system maintenance and performance monitoring.
- Training delivery partners to follow standardized delivery protocols.
- Designing efficient coordination systems between restaurants and delivery partners.

Quality Assurance prevents delays before they occur and ensures long-term service consistency.

6.4 Role of Sampling in Quality-Related Decision Making

Monitoring every delivery transaction is impractical in high-volume digital service environments. Sampling enables efficient performance evaluation using selected observations.

In delivery time analysis, sampling is used to:

- Collect delivery time data from selected orders across different time periods.
- Form rational subgroups for control chart analysis.
- Estimate overall service performance without monitoring every delivery.
- Detect delivery performance trends efficiently.

Sampling supports timely, cost-effective, and reliable decision-making.

6.5 Quality Audits in the Swiggy Delivery Process

Quality audits are systematic reviews of operational performance to ensure compliance with defined standards and procedures.

In the Swiggy delivery system, quality audits help to:

- Evaluate adherence to delivery protocols and service standards.
- Identify inefficiencies in order assignment and partner allocation.
- Verify accuracy of delivery time monitoring systems.
- Ensure corrective actions are implemented effectively.
- Promote accountability and continuous improvement.

Regular audits help maintain consistent service quality.

6.6 Role of ISO 9001 in Maintaining Service Quality

ISO 9001 is an internationally recognized quality management standard that promotes process consistency and continuous improvement.

In the Swiggy delivery process, ISO 9001 supports quality by:

- Standardizing delivery service procedures.
- Establishing measurable performance indicators such as delivery time.
- Encouraging risk-based operational planning.
- Requiring documentation of processes and corrective actions.
- Supporting continuous monitoring and systematic improvement.

Implementation of ISO 9001 ensures that quality management becomes structured, measurable, and sustainable.

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7. Conclusion and Learning Outcomes

The conclusion of a Statistical Quality Control study summarizes the overall findings of the analysis and highlights the effectiveness of quality tools in understanding and improving real-life service processes. Learning outcomes reflect the knowledge and practical understanding gained through the application of statistical methods, data interpretation, and process evaluation. In digital service platforms such as online food delivery, systematic quality analysis

plays an essential role in improving operational efficiency, enhancing customer satisfaction, and supporting continuous service improvement.

Interpretation and Learning Outcomes

1. The project demonstrated the practical application of Statistical Quality Control tools in analyzing the Swiggy food delivery process.
2. Food delivery time was identified as a key quality characteristic affecting service efficiency and customer satisfaction.
3. Quality control tools helped identify major operational factors contributing to delivery delays.
4. Control chart analysis showed that a process can be statistically stable but still not capable of meeting service expectations.
5. Process capability analysis provided quantitative evidence of performance gaps and the need for improvement.
6. The project enhanced practical understanding of data-driven decision-making, continuous monitoring, and process improvement in service operations.

References

- **Fishbone Chart:** Word Doc
- **Data & QC Tools:** Excel File
- **Power BI ETL, Data Cleaning:** Power BI File
- **Kaggle Dataset:** Link